

Bryce Tabangcura - Lab Week 2

Lab partner: haven bunter-gooley

Pre-Lab

1. Equation that relates n , θ , d and λ

$$d \sin(\theta) = n\lambda$$

2. Solving for 4000 and 440

$$d \sin(\theta) = n\lambda$$

$$\theta = \arcsin\left(\frac{n\lambda}{d}\right)$$

$$d = \frac{1 \text{ cm}}{4000} = \frac{0.01 \text{ m}}{4000}$$

$$d = 2.5 \times 10^{-6} \text{ m}$$

$$n = 1$$

$$\lambda = 0.44 \times 10^{-6} \text{ m}$$

$$\theta = \arcsin\left(\frac{1 \times 0.44 \times 10^{-6}}{2.5 \times 10^{-6}}\right)$$

$$\theta = \arcsin(0.176)$$

$$\theta \approx 10.14^\circ$$

Preamble

Experiment 1

Aim

Method

Results

Conclusions